

SHUVAM PAL

+91 85830 54679 ✉ ishuvam.pal@gmail.com  Shuvam Pal  unichronic

Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built failure-aware automation around multi-step public-site workflows with retries, state validation, session recovery, API lookups, and operator checkpoints.
- Optimized an AI-assisted drafting pipeline that generated structured suggestions and carried validated outputs into downstream automation.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python inference workflows inside a mature open-source application, integrating preprocessing, model execution, output validation, and inspectable generated artifacts.
- Reduced processing runtime by **87%** with asynchronous and parallel execution across heavy data paths.
- Worked through reliability issues across large inputs, typed outputs, label mapping, and PyTorch/ONNX-style model integration.

Open Source

Contributions 

- Contributed reliability and systems patches across **Kuadrant MCP Gateway**, Kyverno, CRIU, VideoLAN, and Tiled, including backend-outage grace handling, parser robustness, and security-validation paths.

Projects

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, yfinance, JSONL*

GitHub 

- Built a multi-agent market-research workflow with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, and replayable JSONL traces.
- Evaluated decisions with historical-market backtests, reward-loop style scorecards, validation harnesses, and baseline comparisons across return, drawdown, win-rate, and risk signals.

Hyoka | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, OpenTelemetry, Next.js, Docker*

- Built an **AI-agent reliability control plane** with SDK/proxy/OTLP trace ingestion, evaluator registries, replay modes, failure mining, validation runs, gate decisions, promotions, and audit logs.
- Implemented worker-owned execution leases, content-addressed artifacts, signed manifests, project-scoped API keys, SQLAlchemy/Alembic models, and a Next.js dashboard for operating agent behavior safely.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus*

GitHub 

- Built browser-automation-backed product workflows with REST APIs, RabbitMQ workers, Redis autosave, PostgreSQL persistence, durable job status, and recoverable external publishing tasks.
- Added retries, DLQs, circuit-breaker isolation, health checks, and Prometheus metrics so long-running API/browser workflows stayed observable and fault-tolerant.

Technical Skills

Languages: Python, TypeScript, Go, SQL, JavaScript

Frameworks: FastAPI, Next.js, React, Node.js, Agno, PyTorch, Echo

Tools: PostgreSQL, SQLite, Redis, Vector Memory, OpenTelemetry, Docker, GCP, GitHub Actions, JSONL